

Packing circuits below the corank: a kernel for binary matroids and edge-disjoint bond packing

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Abstract

For a matroid M , let $\nu(M)$ be the maximum number of pairwise element-disjoint circuits and let $\delta(M) = r^*(M) - \nu(M)$, where r^* is the corank. We give a polynomial-time kernelization for deciding $\delta(M) \leq k$ on binary matroids supplied by a binary representation. Its element bound is $2^{O(4^k)}$; this is a double-exponential bound. The argument combines a counting lemma valid for every matroid, an exchange rule for repeated binary circuit profiles, and contraction of series pairs. For simple graphs we also give a direct treatment of edge-disjoint bonds: every two-connected block H on at least three vertices satisfies $\tau(H) \leq 4\delta_{\text{bond}}(H)$. A false-twin surplus rule yields a $2^{O(k)}$ -vertex kernel using independent block copies.

1 Introduction

Every circuit of M contains an element outside any fixed basis, so $\nu(M) \leq r^*(M)$. We study the amount by which the maximum packing falls below this elementary upper bound. For the graphic case, where r^* is the cyclomatic number and circuits are cycles, Harant et al. [4, Theorem 3] proved that for each fixed k the relevant reduced graphs form a finite family, which yields fixed-parameter tractability for this graphic case. The present binary representation permits circuits beyond graphic matroids. Rautenbach and Regen [6] study the distinct vertex-disjoint cycle variant. The dual of a graphic matroid gives a second motivating case: its circuits are graph bonds, and its corank is $|V| - c(G)$. Every nonempty cut contains a bond, so replacing each member of an edge-disjoint cut packing by a contained bond preserves the packing size; the two maximum packing numbers are equal. Cut packing was studied by Caprara, Panconesi, and Rizzi [2].

Put

$$g(k) = 2k + 2^{2k} + 2k2^{2^{2k}}2^{2k} \quad (k \geq 0). \quad (1)$$

In particular $g(k) = 2^{4^k + O(k)}$ for positive k .

Theorem 1 (Binary-matroid kernel). *Given a binary matrix A representing M and $k \in \mathbb{Z}_{\geq 0}$, there is a polynomial-time algorithm, measured in the input encoding length L , that outputs an equivalent instance (A', k') with $k' \leq k$ and at most $\max\{3, g(k)\}$ elements. After row reduction, A' has $O(\max\{3, g(k)\}^2)$ bits. The problem is fixed-parameter tractable. For $k < 0$, the algorithm outputs a fixed NO instance.*

Theorem 2 (Bond packing). *Let G be a finite simple graph, and let $\nu_{\text{bond}}(G)$ be the maximum number of pairwise edge-disjoint bonds. With $\delta_{\text{bond}}(G) = |V(G)| - c(G) - \nu_{\text{bond}}(G)$, deciding $\delta_{\text{bond}}(G) \leq k$ is fixed-parameter tractable and has a $2^{O(k)}$ -vertex kernel. Every two-connected block H on at least three vertices satisfies $\tau(H) \leq 4\delta_{\text{bond}}(H)$.*

The binary-matroid proof does not rely on recognizing an optimal packing: it finds a small set of exceptional nonbasis elements by vertex cover. Since a cographic matroid is binary, the FPT

assertion of Theorem 2 also follows from Theorem 1 via a polynomial-time binary representation. The direct graph proof gives the vertex-cover bound and the smaller vertex kernel: in a YES instance at most k nontrivial blocks survive, each with at most $4k 2^{4k-1}$ vertices.

2 Preliminaries

All matroids are finite. A circuit is a minimal dependent set, and B denotes a basis. Write $B^* = E(M) \setminus B$, so $|B^*| = r^*(M) = |E(M)| - r(M)$. For $e \in B^*$ let $C(e, B)$ denote its fundamental circuit. Loops and coloops are allowed. The binary input representation permits Gaussian elimination, construction of a basis, contraction, and computation of fundamental circuits in polynomial time.

A *cycle* of a binary matroid means a vector of its binary circuit space, identified with its support. This includes the empty cycle. Addition is symmetric difference. The fundamental circuits relative to B form a vector-space basis: each contains exactly one element of B^* , namely its index. Every nonempty cycle is an element-disjoint union of circuits. Contraction by a set projects the cycle space onto the coordinates outside that set. These facts follow as well by writing a binary representation in basis form and taking its nullspace.

We use two elementary reductions. (R1) Delete a coloop. (R2) If $\{x, y\}$ is a two-element cocircuit, contract one member. Both preserve δ . For R1, no circuit uses the coloop and both r^* and ν are unchanged. For R2, binary circuit–cocircuit orthogonality says that every cycle contains x if and only if it contains y . Projection deleting the x coordinate is injective on the cycle space: an element of its kernel would be the singleton cycle $\{x\}$, which cannot contain x without y . Membership of x is recovered from membership of y , so support containment and disjointness are preserved in both directions. The projection therefore bijects circuits and their disjoint packings. It preserves the circuit-space dimension r^* and the optimum ν . The pair may lie inside a larger component; R2 contracts one member.

For a graph, a bond is a minimal nonempty edge cut. Equivalently, in a connected component it is a cut $\partial(S)$ for which both shores induce connected graphs. The two definitions agree: if one shore is disconnected, the edges leaving one of its components form a proper nonempty cut subset; conversely, if both shores are connected, omitting any cut edge reconnects the shores, so no proper subset is a cut. We write $\tau(G)$ for the minimum vertex-cover size. A two-connected block below always has at least three vertices; bridge blocks are treated separately.

3 A counting lemma for arbitrary matroids

Lemma 3. *For any matroid M , any basis B , and any maximum circuit packing $\mathcal{P}$, let q be the number of packed circuits containing at least two members of B^* . Then $q \leq \delta(M)$. The other packed circuits are pairwise disjoint fundamental circuits $D_i = C(e_i, B)$. If $F = B^* \setminus \{e_i\}$, then*

$$|F| = \delta(M) + q \leq 2\delta(M). \quad (2)$$

Proof. A circuit cannot lie in the independent set B , so every packed circuit contains at least one member of B^* . Put $a_C = |C \cap B^*|$. Since the packing is element-disjoint, $\sum_{C \in \mathcal{P}} a_C \leq |B^*| = \nu(M) + \delta(M)$. Since every $a_C \geq 1$ and q of them are at least two, the same sum is at least $\nu(M) + q$. Thus $q \leq \delta(M)$. If $a_C = 1$ with $C \cap B^* = \{e\}$, then $C \subseteq B \cup \{e\}$ and uniqueness of the circuit in $B \cup \{e\}$ gives $C = C(e, B)$. There are $\nu(M) - q$ such circuits, with distinct indexing elements and disjoint supports. Subtracting their indexing elements from B^* gives $|F| = r^*(M) - (\nu(M) - q) = \delta(M) + q$. $\square$

4 An exchange lemma for repeated binary profiles

Fix a basis B of a binary matroid and *any* pairwise disjoint family $D_i = C(e_i, B)$ indexed by I ; this family need not be optimal. Write

$$D_i = \{e_i\} \dot{\cup} P_i, \quad F = B^* \setminus \{e_i : i \in I\}, \quad P_0 = B \setminus \bigcup_{i \in I} P_i,$$

and, for $f \in F$,

$$C(f, B) = \{f\} \dot{\cup} R_f \dot{\cup} \bigsqcup_{i \in I} A_{fi}, \quad R_f \subseteq P_0, \quad A_{fi} \subseteq P_i.$$

For $x \in P_i$ define its full signature $s_i(x) = \{f \in F : x \in A_{fi}\}$. The *profile* of D_i is the multiset of these signatures for $x \in P_i$, with the labels of F fixed. Keeping the entire joint signature, rather than only the individual cardinalities $|A_{fi}|$, is essential.

Lemma 4 (Exchange or H1). *If $T \subseteq I$ consists of t circuits with the same profile and $t > |F|$, then for any $i_0 \in T$,*

$$\nu(M/D_{i_0}) = \nu(M) - 1, \quad \delta(M/D_{i_0}) = \delta(M).$$

Proof. Put $h = |F|$ and $D = D_{i_0}$. First assume $h \geq 1$, so after the contraction there are $s = t - 1 \geq h \geq 1$ retained circuits of type T . Equality of signature multisets supplies a common position set L and bijections $\phi_i : L \rightarrow P_i$ such that the signature at a given position is the same in every P_i . For $Z \subseteq F$ put $a_i(Z) = \Delta_{f \in Z} A_{fi}$. Under ϕ_i , all sets $a_i(Z)$ correspond to the same subset $a(Z) \subseteq L$; in particular they are simultaneously empty or nonempty.

Every cycle X of M has a unique expression

$$X = \Delta_{f \in Z} C(f, B) \Delta \Delta_{i \in I: u_i=1} D_i, \quad Z = X \cap F, \quad u_i = \mathbf{1}_{e_i \in X}. \quad (3)$$

Only the indicated fundamental circuits meet D_i , hence

$$X \cap D_i = \begin{cases} a_i(Z), & u_i = 0, \\ D_i \setminus a_i(Z), & u_i = 1. \end{cases} \quad (4)$$

When $a(Z) \neq \emptyset$, both alternatives in (4) are nonempty: $a_i(Z) \subseteq P_i$ while the other alternative contains e_i . Also $Z \neq \emptyset$. A circuit of this kind meets every T -block and uses at least one member of F ; call it *crossing*. A disjoint packing has at most h crossing circuits, because their nonempty intersections with F are disjoint, and it cannot simultaneously contain a T -block circuit. If $a(Z) = \emptyset$, the circuit meets each T -block in either nothing or the whole block. If it contains a whole block, circuit minimality forces it to equal that block. Otherwise call it *neutral*. Thus crossing, block, and neutral circuits exhaust all circuits.

We next justify that the family of neutral circuits is identical in M and $N = M/D$, as subsets of $E(M) \setminus \bigcup_{i \in T} D_i$. Since D is a circuit, $B' = B \setminus P_{i_0}$ is a basis of N : it has $r(M) - (|D| - 1)$ elements and its image spans the contraction. The fundamental circuits in N are $C(f, B) \setminus D$ for $f \in F$ and D_i for $i \neq i_0$. Therefore (4) still holds on every retained block. Take a circuit of N avoiding all retained blocks. On any one retained block, absence of e_j gives $u_j = 0$, and then (4) gives $a_j(Z) = \emptyset$. Profile equality gives $a_{i_0}(Z) = \emptyset$. Lift its fundamental-circuit expression to M , choosing $u_{i_0} = 0$; the lift has no coordinates on D , so it has precisely the original support. It is a cycle of M . If it contained a smaller nonempty cycle of M , that cycle would avoid D and project to a smaller nonempty cycle of N , contradicting circuit minimality. Hence it is a circuit of M . Conversely, a neutral circuit of M projects without changing its support to a nonempty cycle of N . That cycle contains a circuit of N ; this circuit avoids retained blocks and so, by

the preceding argument, is also a circuit of M . Minimality in M forces equality. The arbitrary sets R_f never affect this argument: all coordinates outside D are retained unchanged.

Let ρ be the maximum packing number of this common neutral family. In M , a packing without crossing circuits has at most t block circuits and ρ neutral circuits; one with crossing circuits has no T -block circuits and at most $h + \rho < t + \rho$ members. The t disjoint block circuits together with a maximum neutral packing attain $t + \rho$, hence $\nu(M) = t + \rho$. In N the same crossing, block, and neutral trichotomy applies to the nonempty retained family $T \setminus \{i_0\}$ of matching profiles. It gives an upper bound $\max\{s, h\} + \rho = s + \rho$, attained by all s retained blocks and a maximum neutral packing. Thus $\nu(N) = s + \rho = \nu(M) - 1$. A circuit has rank $|D| - 1$, so

$$r^*(N) = (|E(M)| - |D|) - (r(M) - (|D| - 1)) = r^*(M) - 1.$$

Subtracting packing numbers proves the second assertion.

If $h = 0$, all fundamental circuits are the disjoint D_i ; basis elements outside them are coloops. Every nonempty cycle is a union of some D_i , and its minimal nonempty supports are exactly the D_i . Contracting one D_i removes precisely that circuit and reduces both r^* and ν by one. This includes $t = 1$ and a singleton-loop block. $\square$

The strict threshold matters. Take the binary matroid on $E = \{0, 1, \dots, 7\}$ with basis $B = \{2, 4, 6, 7\}$ and fundamental circuits $C(0, B) = \{0, 2, 4\}$, $C(1, B) = \{1, 2, 6\}$, $C(3, B) = \{3, 4, 7\}$, and $C(5, B) = \{5, 6, 7\}$. These four circuits specify its binary cycle space. The disjoint block circuits $\{1, 2, 6\}$ and $\{3, 4, 7\}$ have the same profile over $F = \{0, 5\}$, so $t = h = 2$. Contracting either one of these two blocks changes (r^*, ν, δ) from $(4, 2, 2)$ to $(3, 2, 1)$. Thus $t = h$ cannot replace $t > h$.

5 Kernel for binary matroids

Fix any basis B . Make a conflict graph whose vertices are B^* : two vertices e, f are adjacent if $C(e, B)$ and $C(f, B)$ intersect. An independent set is exactly a family of disjoint fundamental circuits; its complement is a vertex cover.

Lemma 5 (A small witness can be found). *If $\delta(M) \leq k$, the conflict graph has a vertex cover of size at most $2k$. In time $O(4^k \text{poly}(L))$ one can find such a cover, or correctly reject $\delta(M) \leq k$.*

Proof. Apply Lemma 3 to a maximum packing relative to the chosen B . Its good fundamental circuits are disjoint and their indexing vertices form an independent set; the complementary set F is a cover of size at most $2\delta(M) \leq 2k$. This proves the existence statement without requiring the algorithm to find the maximum packing. For the algorithm, repeatedly take an uncovered edge ef and branch on putting e or f into the cover. Each branch reduces the remaining budget by one; a search to depth $2k$ has at most $2^{2k+1} - 1$ nodes, and each node is processed in polynomial time. Exhausting all branches proves that no such cover exists and hence that the instance is a NO instance. A found cover is only a witness for subsequent reductions, not proof of YES. $\square$

Given the cover F , let $I = B^* \setminus F$. We apply R1, R2, and Lemma 4 whenever their conditions hold. The exchange rule compares profile multisets and contracts one whole block from any class occurring more than $|F|$ times. R1 removes a zero coordinate column from the cycle-space representation. R2 contracts one member of a pair of equal nonzero columns. Equality of nonzero cycle-space columns is exactly the two-element cocircuit condition: the corresponding coordinates agree on every cycle, so their pair is a cocycle, while neither singleton is a cocycle. When needed, these columns and profiles are obtained by linear algebra and sorting. No exact computation of ν enters the reductions.

The witness can be maintained. R1 deletes a basis element because a nonbasis column in the fundamental-circuit representation is a nonzero unit column. Every R2 pair has at least one basis element, since distinct nonbasis elements have distinct unit columns; choose that basis element for contraction. After either operation, remove it from B and from any affected P_i or P_0 . The remaining fundamental circuits are exactly the old ones with the removed coordinate deleted, remain disjoint, and the labelled set F is unchanged. After H1 contracts $D_i = \{e_i\} \dot{\cup} P_i$, remove P_i from B and i from I ; again F remains unchanged and the other blocks are fundamental circuits. Each operation removes at least one element. Thus there are at most $|E(M)|$ operations and no loop caused by witness recomputation.

Lemma 6 (Terminal size). *If $h = |F| \leq 2k$ and none of R1, R2, H1 applies to the maintained witness, then $|E(M)| \leq g(k)$.*

Proof. Partition $E(M) = F \dot{\cup} P_0 \dot{\cup} \bigsqcup_{i \in I} D_i$. For $x \in P_i$, its coordinate in a cycle with switches $(Z, (u_j))$ is $u_i + |s_i(x) \cap Z| \pmod{2}$; the coordinate of e_i is u_i . For $x \in P_0$ it is $|s_0(x) \cap Z| \pmod{2}$, where $s_0(x) = \{f \in F : x \in R_f\}$. These formulas follow by expanding (3). A member of P_0 with empty signature is a coloop and would be removed by R1. Two members with the same nonempty signature have identical nonzero columns and would trigger R2. Consequently $|P_0| \leq 2^h - 1$.

Within each D_i , e_i supplies the empty signature. A member of P_i with empty signature agrees with e_i on every cycle and would trigger R2. Two members of P_i with the same nonempty signature would do the same. Hence P_i contains at most one representative of each nonempty subset of F and $|D_i| \leq 2^h$. Its profile is now a subset of the $2^h - 1$ possible nonempty signatures, so there are at most $2^{2^h - 1}$ profiles. Since H1 is inapplicable, each profile occurs at most h times. Therefore

$$|E(M)| \leq h + (2^h - 1) + h 2^{2^h - 1} 2^h \leq 2k + 2^{2k} + 2k 2^{2^{2k}} 2^{2k} = g(k).$$

Here g is a convenient upper bound. Blocks consisting of a single loop have the empty profile and are included. If $h = 0$, no block may remain, R1 removes all of P_0 , and the terminal matroid is empty; the displayed bound still holds. $\square$

Proof of Theorem 1. For the FPT algorithm, find a cover using Lemma 5, reject if none exists, then apply safe reductions to a terminal instance. Lemma 4 and the R1/R2 arguments preserve δ at every step, and Lemma 6 bounds the remaining element count by $g(k)$. Enumerating the 2^{r^*} cycles of the terminal binary representation, retaining minimal nonzero supports, and checking all subsets of those supports for pairwise disjointness computes ν exactly in a function of k times a polynomial in the input size. It decides the original instance, including the case where a small cover exists but $\delta > k$.

For a *standard polynomial-time kernelization*, one further wraps this FPT preprocessing. If $k < 0$, return a fixed NO instance represented by $(U_{1,3}, 0)$. For $k \geq 0$ and $n = |E(M)| \leq g(k)$, row-reduce the representation to full row rank and return it with $k' = k$. Otherwise run the cover search and reductions. Since $n > g(k) \geq 4^k$ for $k \geq 1$, the single $4^k \text{poly}(L)$ cover search costs polynomial time in the original input length; there are at most n reduction steps, each polynomial in L . For $k = 0$, $g(0) = 1$ and the search has constant branching. When the current instance reaches $g(k)$ elements it may be row-reduced and returned with $k' = k$. If the search finds no cover, the branch is NO by Lemma 5; encode it as $(U_{1,3}, 0)$. This matroid has three parallel elements, $r^* = 2$, $\nu = 1$, and $\delta = 1 > 0$. Its three elements explain the $\max\{3, g(k)\}$ bound in the statement. The comparison with $g(k)$ is carried out with integers saturated at $n + 1$, so an enormous value of $g(k)$ is never explicitly constructed. In every return branch the output matrix has full row rank and at most $O(\max\{3, g(k)\}^2)$ bits, since its rank is at most its number of columns. Thus the wrapper is polynomial-time and gives a kernel; the exhaustive terminal solver proves FPT. $\square$

6 Edge-disjoint bond packing

We give a graph-specific route to Theorem 2. We use independent copies of blocks: an articulation vertex shared by two blocks appears once in each block instance.

Lemma 7 (Block additivity). *The bonds of a graph are precisely the bonds of its blocks, regarded as edge sets of the original graph. In particular, with bridge blocks included,*

$$\nu_{\text{bond}}(G) = \sum_H \nu_{\text{bond}}(H), \quad |V(G)| - c(G) = \sum_H (|V(H)| - 1), \quad \delta_{\text{bond}}(G) = \sum_H \delta_{\text{bond}}(H).$$

Every bridge block has deficit zero.

Proof. It suffices to work in a connected component. The block-cut incidence graph is a tree. A simple cycle cannot enter another block through an articulation vertex and return without repeating that vertex. Hence each cycle lies within one block, and the graphic cycle space is the direct sum of the cycle spaces of the blocks, on disjoint edge sets. Taking orthogonal complements over $\mathbb{F}_2$, the cut space is likewise the direct sum of the block cut spaces. A bond is a nonzero cut-space support minimal under inclusion: a nonminimal cut-space support contains a smaller nonzero cut support, and every nonzero cut support contains a minimal one. Minimal nonzero supports of a direct sum on disjoint coordinate sets belong to exactly one summand, proving the bond assertion. Edge disjoint packings then split and combine blockwise. The rank identity is obtained by adding each block to the block-cut tree: its first articulation vertex has already been counted and its remaining $|V(H)| - 1$ vertices are new. A bridge block has one edge, itself a bond, so both its rank contribution and packing contribution are one. Summing gives the three formulas. $\square$

Lemma 8 (Vertex-cover bound). *If H is a simple two-connected graph on at least three vertices, then $\tau(H) \leq 4\delta_{\text{bond}}(H)$.*

Proof. Let $d = \delta_{\text{bond}}(H)$, and choose a maximum disjoint bond packing $\mathcal{Q}$ of size $|V(H)| - 1 - d$. Fix a spanning tree T . Each bond meets T , for otherwise all its edges could be removed without disconnecting T . If q packed bonds meet T in at least two edges, then the disjointness of the packing gives

$$|V(H)| - 1 \geq \sum_{Q \in \mathcal{Q}} |Q \cap E(T)| \geq |\mathcal{Q}| + q = |V(H)| - 1 - d + q;$$

thus $q \leq d$. A packed bond meeting T in exactly one edge e is the fundamental cut of e : no other tree edge crosses the cut, so each component of $T - e$ lies in one shore. Call these $|\mathcal{Q}| - q$ tree edges good, and let F_T consist of the remaining tree edges. Then $|F_T| = |V(H)| - 1 - (|\mathcal{Q}| - q) = d + q \leq 2d$.

The tree path of a graph edge can contain at most one good edge. Indeed, an edge whose path contains a good edge crosses its fundamental cut and belongs to the corresponding packed bond; two good edges on the path would place the graph edge in two disjoint packed bonds. Contract each component of the forest $(V(H), F_T)$ to a vertex. The uncontracted good edges form a tree Q . Every graph edge either remains inside one contracted vertex or joins adjacent vertices of Q , by the preceding path property.

Consider a contracted vertex consisting of a single original vertex v ; it is not incident with F_T . If it had degree at least two in Q , deleting v would separate the corresponding branches of Q , and no graph edge could reconnect them, since graph edges join only equal or adjacent Q vertices. This contradicts two-connectivity. Hence every singleton contracted vertex has Q -degree at most one. If two singleton contracted vertices were adjacent by an edge of H , they would be adjacent in Q . Since both have degree at most one, Q would consist of just these two vertices; as all vertices of Q are singletons, H would have only two vertices. This is excluded. Therefore every edge of H has an endpoint incident with F_T . The endpoints of F_T form a vertex cover of size at most $2|F_T| \leq 4d$. $\square$

Lemma 9 (False-twin surplus). *Let H be a simple connected graph, and let T be an independent set of t vertices with the same open neighborhood N , of size $d \geq 1$. If $t > d$, deleting any one $v \in T$ preserves $\delta_{\text{bond}}(H)$.*

Proof. For $x \in T$, put $D_x = \partial(\{x\})$. There is another member of T , and any path segment $a - x - b$ with $a, b \in N$ can be rerouted as $a - y - b$ through such a member y . Hence $H - x$ is connected, and D_x is a bond. The D_x are edge-disjoint and each has d edges.

Let $Q = \partial(S)$ be a bond meeting D_x but unequal to it, with $x \in S$. If all of N lay in S , Q would not meet D_x . If all of N lay outside S , then x would be isolated in $H[S]$; connectivity of that shore would force $S = \{x\}$ and $Q = D_x$. Thus each shore contains a vertex of N , and Q meets every D_y , $y \in T$. Call Q crossing. In an edge-disjoint packing there are at most d crossing bonds, because each uses a distinct edge of any fixed star D_x . A crossing bond prevents the presence of any D_y . Replacing the crossing bonds by all $t > d$ stars strictly increases the packing. If there are no crossing bonds, the packing may be enlarged to contain every star, because every other packed bond avoids all star edges. Consequently some maximum packing contains all t stars.

Let ρ be the largest packing of bonds avoiding every star edge. A cut avoiding all star edges keeps $T \cup N$ on one shore. Deleting v from that shore preserves its connectivity: occurrences of v on a path can be rerouted through a remaining twin, and the other shore is unchanged. Conversely, adding v to the shore containing $T \setminus \{v\}$ and N preserves both shores' connectivity. The cut edge set itself is unchanged. Thus these neutral bonds, including their edge-disjointness relations, correspond in H and $H - v$, and have the same packing number ρ . We have $\nu_{\text{bond}}(H) = t + \rho$.

In $H - v$ the twin class has $t - 1 \geq d$ members. Its stars are still bonds: when $t - 1 \geq 2$ the same rerouting proves this, while the only case $t - 1 = 1$ has $d = 1$ and its star is a bridge. The same crossing argument bounds their number by d . Replacing crossing bonds by all $t - 1$ stars now does not decrease packing size, even at equality $t - 1 = d$. Hence $\nu_{\text{bond}}(H - v) = t - 1 + \rho$. The path rerouting also shows $H - v$ remains connected, so its rank contribution $|V| - 1$ decreases by one. The deficit is unchanged. $\square$

For isolated false twins, deleting a vertex decreases both $|V|$ and $c(G)$ by one and changes no bond; its deficit is unchanged.

Lemma 10 (Size of a reduced block). *Let H be a two-connected simple block on at least three vertices, and suppose no false-twin class in this block instance has size larger than its neighborhood. Then $|V(H)| \leq \tau(H)2^{\tau(H)-1}$.*

Proof. Take a vertex cover C of size $\tau = \tau(H)$, and put $I = V(H) \setminus C$. The set I is independent. A two-connected graph on at least three vertices has minimum degree at least two, so every $x \in I$ has a neighborhood $S \subseteq C$ with $|S| \geq 2$. For each such S , the vertices in I with neighborhood S are contained in a false-twin class, which by assumption has at most $|S|$ members. These groups partition I ; hence

$$|V(H)| \leq \tau + \sum_{S \subseteq C, |S| \geq 2} |S| = \tau + (\tau 2^{\tau-1} - \tau) = \tau 2^{\tau-1}.$$

The identity in the middle counts each vertex of C in exactly $2^{\tau-1}$ subsets and subtracts the τ singleton subsets. $\square$

Proof of Theorem 2. Use Lemma 7 to split G into independent block instances, with separate copies of every articulation vertex. Bridge and isolated-vertex instances have zero deficit and may be discarded. Within a current block, apply Lemma 9; if a deletion destroys two-connectivity, replace the block by its new independent block instances. The deficit is unchanged by the surplus rule and additive under the replacement. Every nontrivial deletion reduces the

sum of block rank contributions $\sum_H(|V(H)| - 1)$ by one; splitting preserves that sum. Thus at most $|V(G)| - c(G)$ nontrivial deletions occur. Block decomposition, twin grouping, and the graph updates take polynomial time per step.

If $k < 0$, reject. If $k = 0$, accept exactly when no nontrivial block remains: each such block has $\tau \geq 2$ and Lemma 8 forces positive integral deficit. For $k \geq 1$, if more than k nontrivial blocks remain, reject by additivity and the same positive-deficit observation. Otherwise every surviving block in a YES instance has deficit at most k . Lemmas 8 and 10 therefore bound its size by $N(k) = 4k 2^{4k-1}$. If any reduced block exceeds this size, reject. For a kernel, output the disjoint union of the at most k surviving nontrivial block instances, with parameter k . Additivity preserves the deficit, and the union has at most $kN(k) = 2^{O(k)}$ vertices. A rejected branch outputs the fixed NO instance $(K_3, 0)$; when no nontrivial block remains, output the empty graph with parameter 0. The reductions and block construction take polynomial time, with size comparisons saturated at the input vertex count, so this is a vertex kernel.

For the FPT decision algorithm, on the at most k remaining blocks, enumerate bipartitions and retain cuts with connected shores. A dynamic program whose state is the set of edges already used either skips a bond or adds it when its edge set is disjoint from the state. It finds a maximum edge-disjoint packing. Each block has at most $\binom{N(k)}{2}$ edges and at most $2^{N(k)-1}$ distinct shores to inspect, so the dynamic program takes $2^{O(N(k)^2)} \text{poly}(|V(G)|)$ time. Sum the exact block deficits and compare with k . All rejection tests are necessary for a YES instance, and every other step preserves or exactly computes the deficit; this proves the FPT assertion. $\square$

The independent-copy qualification is material. In the graph with edges 02, 03, 04, 12, 13, 14, 25, 56, 62, a $K_{2,3}$ block and a triangle meet at vertex 2. Vertex 2 belongs to a false-twin class within the bipartite block, but has additional neighbors in the whole graph. Deleting it from the whole graph changes the deficit from 2 to 1, whereas reducing only its copy inside the bipartite block preserves both block deficits.

7 Computational verification

Scripts, summary data, and reproduction commands are available at <https://github.com/zhourong-bot/binary-matroid-circuit-packing> (archived at Zenodo, doi:10.5281/zenodo.22952617).

The binary-matroid checks processed the complete catalogue of 3,766 isomorphism classes on at most ten elements and its 134,860 bases, as well as constructed and random inputs with at most sixteen elements and selected cographic $K_{a,b}$ instances with up to 32 elements. The kernel pipeline was checked on 4,021 inputs, each at $k = \delta$ and $k = \delta - 1$, for 8,042 parameter choices. The input deficits 0, 1, 2, 3, 4, 5, 6 occurred respectively 478, 1264, 1633, 564, 54, 21, 7 times. There were 15,716 local checks: 14,344 pipeline reductions and 1,372 nonminimum-cover H1 targets, with no observed deficit failure. Among the terminal instances with $\delta = 1, 2, 3$, the maximum element counts were respectively 9, 13, 20. The implementation used brute-force cover enumeration on these small inputs; this does not benchmark the bounded search tree of Theorem 1.

The graph-side enumeration covered all two-connected simple graphs on at most nine vertices (201,727 graphs). A separate check of the cover argument processed 70 two-connected graphs and 9,588 spanning trees. These finite checks test the implementations; they are not proofs of the lemmas or bounds.

8 Related work

For graphic matroids, Harant et al. [4] already give finite obstruction sets for each deficit k and fixed-parameter tractability for edge-disjoint cycle packing below the cyclomatic number.

Rautenbach and Regen [6] treat the vertex-disjoint cycle variant. Theorem 1 covers explicitly represented binary matroids. For bonds, the cographic matroid $M^*(G)$ is binary and has $r^*(M^*(G)) = r(M(G)) = |V(G)| - c(G)$ and $\nu(M^*(G)) = \nu_{\text{bond}}(G)$. Hence its FPT conclusion follows from Theorem 1. The graph-specific contribution is the direct $\tau \leq 4\delta_{\text{bond}}$ proof and the $2^{O(k)}$ -vertex kernel with at most k nontrivial blocks of at most $N(k) = 4k2^{4k-1}$ vertices each.

Caprara, Panconesi, and Rizzi [2] study packing cuts. Since each nonempty cut contains a bond, their cut-packing setting and edge-disjoint bond packing have the same maximum packing number. Gutin and Mnich [3] survey parameterization above and below guaranteed values. Related matroid packing and covering problems appear in [1, 5, 7].

9 Open problems

The element bound $2^{O(4^k)}$ leaves open whether the binary-matroid problem has a single-exponential or polynomial kernel. For two-connected graph blocks, the proved cover bound is $\tau \leq 4\delta_{\text{bond}}$; enumeration through nine vertices found no counterexample to $\tau \leq 2\delta_{\text{bond}}$, for which we give no proof. A direction not treated here is profile exchange and compression for matroids representable over $GF(q)$ with $q > 2$; the binary symmetric-difference argument above does not establish that case.

AI-use disclosure

Multiple AI systems assisted with candidate arguments, proof review, finite-instance checks, and drafting. The author takes responsibility for the manuscript.

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